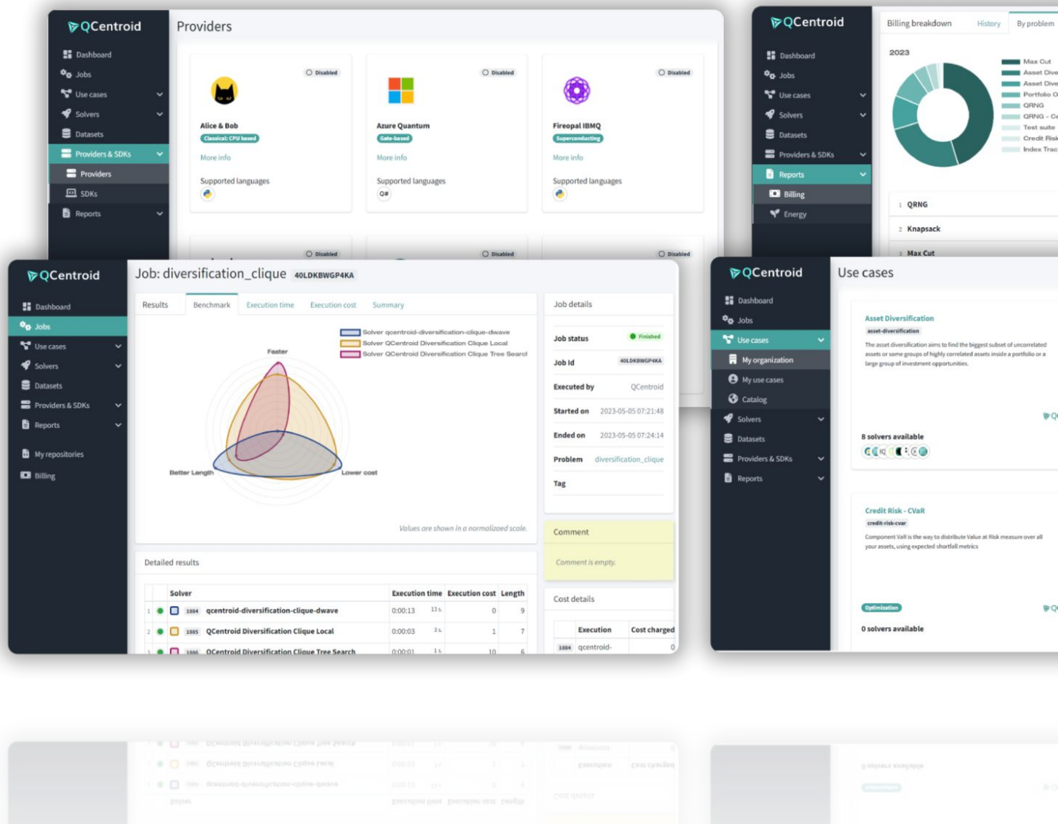
QCentroid

Get started now!

The platform to adopt Quantum Computing without complexity



QCentroid

ETH zürich

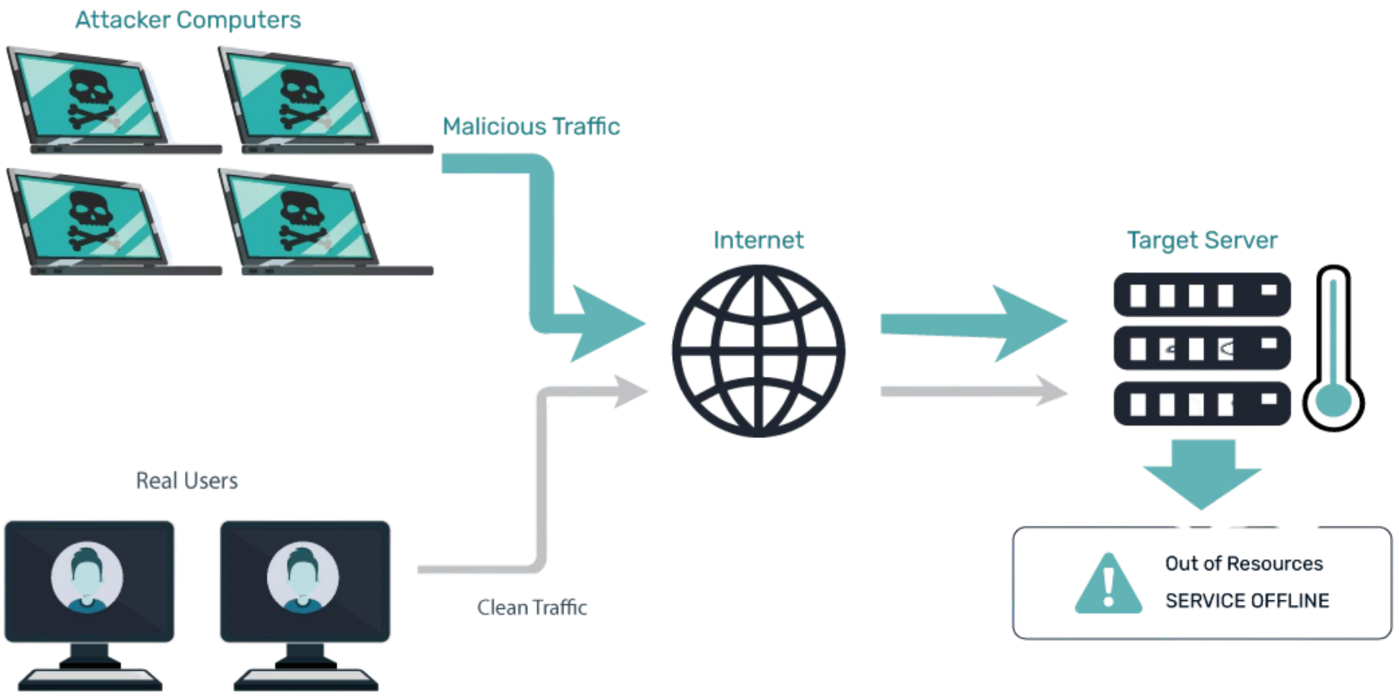
DDoS Anomaly Detection in Network Traffic

Using Quantum-Inspired Machine Learning

Co-developed with GSMA™

Distributed Denial of Service (DDoS)

Background



Detect DDoS attacks in real network traffic

The challenge

QCentroid

University Campus Network Data

Real flow-level telemetry, 100k rows per 15-min window.

Real attack patterns

Engineered DDoS bursts seeded into live-style traffic.

Real industry problem

Co-developed with GSMA - used in their Quantum Telco Playbook

Co-developed with GSMA™

What happens if you win

Built for the weekend, seen by the industry

- **Featured publicly**
  In the post-event write-up on the GSMA Quantum Technologies page.

- **Telco reach**
  750+ operators and 300+ ecosystem members worldwide.

- **In the platform**
  Full solution lives on in the QCentroid × GSMA Discovery Sandbox
  — accessible to GSMA members.

- **Possible Case Study**
  If sufficiently distinct, included as an Alternative Approach in the upcoming GSMA × QCentroid Quantum Telco Playbook.

Co-developed with GSMA

Given test datasets for network flows…

Your task

Identify the time windows containing DDoS activity

Short intervals of consecutive flows
where attack patterns emerge

Identify the malicious source IPs within those windows

Single rows can´t tell you. The signature
is collective, across flows.

Your solution must use Quantum-Inspired Machine Learning

Apply it to the whole pipeline or to a specific substak – what matters is that you can justify your choice

Everything to start building immediately

What you get

QCentroid

Real flow-level data

100k rows x 15-min windows
Two dataset families, evaluated separately

Starter notebook

Data loading, schema, instructions, and submission format – included

QCentroid platform

Simulators + quantum backend
Use it if it suits you or work locally

Full technical detail

Shared with teams who select this challenge

Co-developed with GSMA

Type of solution & justification beat raw scores

Scoring

Criteria 1

Appropriateness

Is the quantum technique a sensible fit for the task you applied it to?

Criteria 2

Justification

Why this technique over the alternatives?

Criteria 3

Realism

Does it plausibly offer an advantage given simulator / hardware constraints?

Criteria 4

Implementation

Clean code, sound experimental setup, rigorous validation

Co-developed with GSMA

What to submit, when, where

Logistics

Your code
Repository or zipped project

A short writeup
Approach - sub-task - rationale - what you observed

Slides to DoraHacks
Same channel as other hackathon challenges

Live demo
Short presentation, depending on final timing

The team behind the challenge

Team

Antonio Peris

Chief Innovation Officer

Alberto Calvo

Chief Technology Officer

Miquel Farre

VP of Engineering

Héctor López

Associate & ETH Student

Co-developed with GSMA™

Build something
we have never seen before

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